

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Dot .Net Core and Progressive Web Application Practical

Practical – IV

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Part A - Theory:

1. What is ASP.NET Web Forms?

ASP.NET Web Forms is a web application framework developed by Microsoft for building dynamic, interactive, and data-driven websites. It provides a rich collection of **server controls** that simplify web application development by reducing the amount of code required.

Features of ASP.NET Web Forms:

- Event-driven programming model.
- Rich collection of server controls.
- Built-in state management using ViewState.
- Easy integration with databases.
- Support for validation controls.
- Rapid application development.

2. What are Server Controls?

Server Controls are controls that run on the server and generate HTML dynamically. They simplify user interaction and automatically handle client-server communication.

Syntax:

```
<asp:ControlName ID="ControlID" runat="server" />
```

3. Common ASP.NET Server Controls

a) Label Control

The **Label** control is used to display static text or messages on a web page.

Syntax

```
<asp:Label ID="lblName" runat="server" Text="Student Name"></asp:Label>
```

b) TextBox Control

The **TextBox** control allows the user to enter text.

Syntax

```
<asp:TextBox ID="txtName" runat="server"></asp:TextBox>
```

c) RadioButton Control

The **RadioButton** control allows the user to select only one option from a group.

Syntax

```
<asp:RadioButton ID="rbMale" runat="server" Text="Male" GroupName="Gender" />
```

d) DropDownList Control

The **DropDownList** control displays a list of options from which the user can select one.

Syntax

```
<asp:DropDownList ID="ddlCourse" runat="server">
  <asp:ListItem>BSc IT</asp:ListItem>
  <asp:ListItem>BSc CS</asp:ListItem>
  <asp:ListItem>BCA</asp:ListItem>
</asp:DropDownList>
```

e) CheckBox Control

The **CheckBox** control allows the user to select or deselect an option.

Syntax

```
<asp:CheckBox ID="chkTerms" runat="server" Text="I Agree" />
```

f) Calendar Control

The **Calendar** control allows users to select a date from a graphical calendar.

Syntax

```
<asp:Calendar ID="Calendar1" runat="server"></asp:Calendar>
```

g) Button Control

The **Button** control is used to perform an action when clicked.

Syntax

```
<asp:Button ID="btnSubmit" runat="server" Text="Submit" OnClick="btnSubmit_Click" />
```

4. Validation Controls

Validation controls are used to ensure that the user enters valid data before submitting a form.

Types of Validation Controls

a) RequiredFieldValidator

Ensures that a control is not left empty.

```
<asp:RequiredFieldValidator
ID="rfvName"
runat="server"
ControlToValidate="txtName"
ErrorMessage="Name is required">
</asp:RequiredFieldValidator>
```

b) RangeValidator

Checks whether a value lies within a specified range.

c) CompareValidator

Compares the value of one control with another control or a fixed value.

d) RegularExpressionValidator

Validates the input using a regular expression (e.g., Email, Mobile Number).

e) CustomValidator

Allows custom validation logic written by the programmer.

f) ValidationSummary

Displays all validation error messages in one place.

5. AdRotator Control

The **AdRotator Control** displays different advertisements or images randomly from an XML file. It is commonly used to display advertisements, announcements, offers, or promotional banners.

Advantages

- Displays advertisements automatically.
- Randomly rotates images.
- Reads advertisement details from an XML file.
- Easy to maintain and update.

Syntax

```
<asp:AdRotator  
ID="AdRotator1"  
runat="server"  
AdvertisementFile="Ads.xml">  
</asp:AdRotator>
```

6. Advantages of ASP.NET Server Controls

- Easy to use and configure.
- Supports event-driven programming.
- Reduces coding effort.
- Provides automatic state management.
- Supports built-in validation.
- Improves application security.
- Generates browser-compatible HTML automatically.

7. Applications

- Student Registration System
- Online Admission Portal
- Employee Registration System
- Online Examination System
- Library Management System
- E-Commerce Registration Forms
- Hospital Appointment System

Part B - Program

- a) Create an ASP.NET Web Forms application using simple server controls.
- a. Design a Student Registration Form using the following ASP.NET controls:
- i. Label and TextBox to accept Student Name
 - ii. RadioButton to select Gender (Male/Female)
 - iii. DropDownList to select Course (BSc IT, BSc CS, BCA)
 - iv. CheckBox for accepting Terms & Conditions
 - v. Calendar control to select Date
 - vi. Button control to submit the form
 - vii. Label control to display the output

Algorithm:

Step 1: Start the program.

Step 2: Create the registration form using [ASP.NET](#) server controls.

Step 3: Accept user inputs (Name, Gender, Course, Date, etc.).

Step 4: Click the **Submit** button to display the entered details.

Step 5: Stop the program.

Code:

CSS:

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;
public partial class _Default : Page
{
    protected void Page_Load(object sender, EventArgs e)
    {
    }
    protected void CheckBox1_CheckedChanged(object sender, EventArgs e)
    {
    }
    protected void Button1_Click(object sender, EventArgs e)
    {
        string name = TextBox1.Text;
        string gender = RadioButtonList1.Text;
        string calender = Calendar1.SelectedDate.ToString();
        string course = DropDownList2.Text;
        Label4.Text= name + "<br/>" + gender + "<br/>" + course + "<br/>" + calender;
    }
}
<%@ Page Title="Home Page" Language="C#" MasterPageFile="~/Site.Master" AutoEventWireup="true"
CodeFile="Default.aspx.cs" Inherits="_Default" %>
<asp:Content ID="BodyContent" ContentPlaceHolderID="MainContent" runat="server">
    <div>
        Student Name<asp:TextBox ID="TextBox1" runat="server"></asp:TextBox>
    </div>
    <br />
    <asp:Label ID="Label2" runat="server" Text="Gender"></asp:Label>
    <asp:RadioButtonList ID="RadioButtonList1" runat="server">
        <asp:ListItem>Female</asp:ListItem>
```

```

        <asp:ListItem>Male</asp:ListItem>
        <asp:ListItem>Other</asp:ListItem>
    </asp:RadioButtonList>
    <br />
    <asp:Label ID="Label3" runat="server" Text="Calender"></asp:Label>
    <asp:Calendar ID="Calendar1" runat="server"></asp:Calendar>
    <br />
    <asp:Label ID="Label1" runat="server" Text="Course"></asp:Label>
    &nbsp;<asp:DropDownList ID="DropDownList2" runat="server">
        <asp:ListItem>B.SC.IT</asp:ListItem>
        <asp:ListItem>B.A.M.M.C</asp:ListItem>
        <asp:ListItem>B.SC.CS</asp:ListItem>
        <asp:ListItem>B.TECH</asp:ListItem>
    </asp:DropDownList>
    <br />
    <br />
    <asp:CheckBox ID="CheckBox1" runat="server" OnCheckedChanged="CheckBox1_CheckedChanged" Text="I agree
with terms and conditions" />
    <br />
    <br />
    <asp:Button ID="Button1" runat="server" Text="SUBMIT" OnClick="Button1_Click" />
    <br />
    <asp:Label ID="Label4" runat="server" Text="Label"></asp:Label>
</asp:Content>

```

Output:

Application name
Home
About

Student Name

Gender
☒ Female
☐ Male
☐ Other

Calender

July 2026						
Sun	Mon	Tue	Wed	Thu	Fri	Sat
28	29	30	1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	1
2	3	4	5	6	7	8

Course

☒ agree with terms and conditions

Akansha
Female
B.SC.IT
16-07-2026 00:00:00

MainContent (Custom) div

Student Name

Gender

☐ Female
☐ Male
☐ Other

Calender

< July 2026 >

Sun	Mon	Tue	Wed	Thu	Fri	Sat
28	29	30	1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	1
2	3	4	5	6	7	8

Course

B.SC.IT

☐ I agree with terms and conditions

SUBMIT

b) Create a Registration form to demonstrate use of various Validation controls.

Algorithm:

Step 1: Start the program.

Step 2: Create the registration form with input fields.

Step 3: Apply validation controls (RequiredFieldValidator, RangeValidator, CompareValidator).

Step 4: Validate the entered data when the button is clicked.

Step 5: Stop the program.

Code:

Default2.aspx:

```

<div>
    <asp:Label ID="Label1" runat="server" Text="Student Name" ValidateRequestMode="Enabled"></asp:Label>
    <asp:TextBox ID="TextBox1" runat="server"></asp:TextBox>
    <asp:RequiredFieldValidator ID="RequiredFieldValidator1" runat="server" ControlToValidate="TextBox1"
ErrorMessage="Name should not empty"></asp:RequiredFieldValidator>
    <br />
</div>
<p>
    <asp:Label ID="Label2" runat="server" Text="Age"></asp:Label>
    <asp:TextBox ID="TextBox2" runat="server" OnTextChanged="TextBox2_TextChanged"></asp:TextBox>
    <asp:RangeValidator ID="RangeValidator1" runat="server" ControlToValidate="TextBox2"
ErrorMessage="Age should be 18<=99"></asp:RangeValidator>
</p>
<p>
    <asp:Label ID="Label5" runat="server" Text="Address"></asp:Label>
    <asp:TextBox ID="TextBox5" runat="server" TextMode="MultiLine"></asp:TextBox>
</p>
<p>
    <asp:Label ID="Label6" runat="server" Text="Email"></asp:Label>
    <asp:TextBox ID="TextBox6" runat="server"></asp:TextBox>
</p>
<p>
    <asp:Label ID="Label3" runat="server" Text="Paaword"></asp:Label>
    <asp:TextBox ID="TextBox3" runat="server" OnTextChanged="TextBox3_TextChanged"
TextMode="Password"></asp:TextBox>
</p>

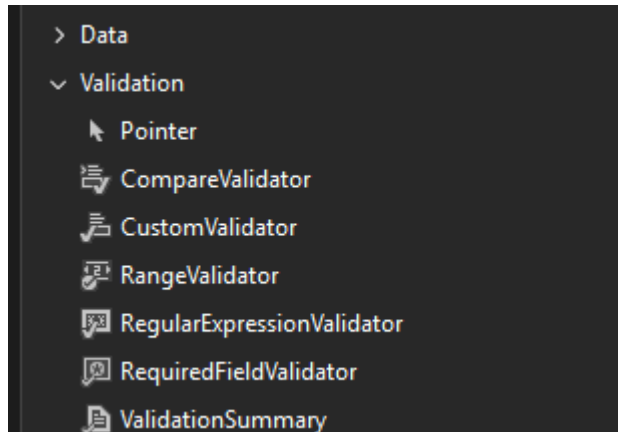
```

```

<asp:Label ID="Label4" runat="server" Text="Confirm Password"></asp:Label>
<asp:TextBox ID="TextBox4" runat="server"></asp:TextBox>
<asp:CompareValidator ID="CompareValidator1" runat="server" ControlToCompare="TextBox3"
ControlToValidate="TextBox4" ErrorMessage="CompareValidator">Password not matched</asp:CompareValidator>
</p>
<p>
<asp:Button ID="Button1" runat="server" Text="Button" />
</p>
</form>
</body>
</html>

```

Output:



BorderWidth	
CssClass	
Display	Static
ErrorMessage	Name should not empty
Font	
ForeColor	
Text	
Behavior	
ClientIDMode	Inherit
ControlToValidate	TextBox1
EnableClientScript	True
Enabled	True

BorderStyle	None
BorderWidth	
CssClass	
Display	Static
ErrorMessage	Age should be 18<=99
Font	
ForeColor	
Text	
Behavior	
ClientIDMode	Inherit
ControlToValidate	TextBox2
CultureInvariantValues	False
EnableClientScript	True

EnableTheming	True
EnableViewState	True
MaxLength	0
ReadOnly	False
Rows	0
SkinID	
TextMode	Multiline
ToolTip	

MaxLength	0
ReadOnly	False
Rows	0
SkinID	
TextMode	Password
ToolTip	

CssClass	
Display	Static
ErrorMessage	CompareValidator
Font	
ForeColor	
Text	Password not matched
Behavior	
ClientIDMode	Inherit
ControlToCompare	TextBox3
ControlToValidate	TextBox4
CultureInvariantValues	False
EnableClientScript	True
Enabled	True

Student Name Name should not empty

Age Age should be 18<=99

Address

Email

Paaword

Confirm Password Password not matched

c) Create Web Form to demonstrate use of Adrotator Control.

Algorithm:

Step 1: Start the program.

Step 2: Create an advertisement XML file with image and URL details.

Step 3: Add the AdRotator control to the web form.

Step 4: Display the advertisement and redirect to the specified URL on click.

Step 5: Stop the program.

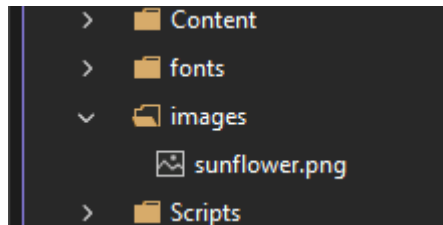
Code:

```
<?xml version="1.0" encoding="utf-8" ?>
<Advertisements>
  <Ad>
    <ImageUrl>images/sunflower.png</ImageUrl>
    <NavigateUrl>https://www.google.com</NavigateUrl>
    <AlternateText>Sunflower</AlternateText>
    <Height>100</Height>
    <Width>100</Width>
  </Ad>
</Advertisements>
```

Default2.aspx:

```
<p>
  <asp:AdRotator ID="AdRotator1" runat="server" AdvertisementFile="~/XMLFile.xml" />
</p>
```

Output:



Student Name

Age

Address

Email

Paaword

Confirm Password



Part C – Curiosity Question

1. Online College Admission System

Develop an **ASP.NET Web Forms application** for an **Online College Admission System**. The application should allow students to register by entering their personal and academic details using appropriate **server controls**. Validate all mandatory fields using suitable **Validation Controls** and display meaningful error messages. After successful registration, display a confirmation message with the entered details. Additionally, use an **AdRotator Control** to display rotating banners for college events, admissions, scholarships, or placement drives.